

# AHLAAM ABDALLAH

Computer Science Graduate | Technology, Society & Development | Human-Centered Computing

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## PROFILE

Computer Science graduate from Kwame Nkrumah University of Science and Technology with experience spanning software development, applied artificial intelligence, cybersecurity, teaching, and community-based education. Interested in how technology can be designed, communicated, and deployed to address real-world challenges in developing communities. Experience building technology systems, evaluating software usability, mentoring students, and supporting educational and community initiatives has developed a strong interest in the relationship between technological innovation, human needs, access, trust, and social development. Particularly interested in technology for education, digital inclusion, cybersecurity awareness, responsible AI, and human-centered technology in Ghana and other African contexts.

## EDUCATION

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**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY (KNUST)** 2022 – 2025

**B.Sc. Computer Science** | Kumasi, Ghana

**Second Class Upper Division**

*Final-Year Project: PetanqueAI — Tournament Management and Prediction System*

Relevant coursework: Artificial Intelligence, Machine Learning, Human-Computer Interaction, Computer Networks, Databases, Software Development

## PROFESSIONAL & TEACHING EXPERIENCE

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**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY (KNUST)** Nov 2025 – Oct 2026

**Teaching Assistant** | Kumasi, Ghana

- Supported undergraduate instruction and student learning in Computer Science coursework, working with students with varied technical backgrounds and learning needs.
- Explained technical concepts in accessible ways, strengthening the ability to translate complex technological ideas for non-specialist audiences.
- Mentored and guided students through coursework, problem-solving, and academic challenges.
- Contributed to final-year project supervision and supported students developing technology-based solutions to real-world problems.

**KTECHHUB** Oct – Dec 2023

**Web Development Intern** | Ghana

- Assisted in developing and maintaining responsive web applications using HTML, CSS, and JavaScript.
- Collaborated with senior developers to implement and refine web-based functionality.
- Gained practical experience translating user requirements into functional digital interfaces.

**ADROIT BUREAU LIMITED** Sep – Nov 2022

**Software Intern** | Ghana

- Tested and evaluated a software application for performance, usability, and user experience.
- Developed a structured book template using the application and provided user-oriented feedback.
- Contributed to product evaluation from the perspective of how effectively technology served its intended users.

## SELECTED TECHNOLOGY PROJECTS

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**PETANQUEAI** 2025

*Tournament Management & Prediction System | KNUST*

- Designed and developed a full-stack tournament management platform addressing practical challenges in organizing and tracking competitive sporting events.
- Built functionality for tournament creation, team management, scoring, match progression, and multiple tournament formats.
- Developed a machine-learning component using structured competition data to generate predictive scores.

- Worked across Python, Flask/FastAPI, Next.js, MongoDB, PostgreSQL, and SQLAlchemy.
- Integrated software engineering, data management, and applied AI to translate a real-world problem into a usable digital system.

## **CYBERSECURITY & DIGITAL SAFETY**

### **Applied Cybersecurity Laboratory Practice**

*TryHackMe | EchoCTF | Kioptrix | Independent SecureShop Lab*

- Conducted practical exercises in network reconnaissance, vulnerability analysis, web security, SQL injection testing, and Linux-based security environments.
- Investigated web-application vulnerabilities through controlled laboratory environments and CTF challenges, including a case study of SQLPad v6.9.0 (CVE-2022-0944) in an EchoCTF environment.
- Used Kali Linux, Nmap, Burp Suite, and related security tools to examine weaknesses in digital systems.
- Developed an increasing interest in the relationship between technical security, user behavior, digital trust, and accessible cybersecurity education.

## **LEADERSHIP & COMMUNITY ENGAGEMENT**

### **MOBILE QUR'AN INSTITUTE (MQI), KNUST**

2021 – 2025

#### **Secretary**

- Coordinated organizational communication, meeting documentation, member records, and administrative activities for a student-led educational institution.
- Prepared meeting minutes and communicated decisions between leadership and members.
- Supported Qur'anic education, seminars, competitions, outreach activities, and community programs.
- Assisted with logistics and coordination across educational and community initiatives.

#### **Additional Leadership**

Most Influential Ladies' Wing Leader, GMSA-KNUST · Secretary, Imaamate Council, GMSA-KNUST · Debate Competition Participant, GMSA-KNUST ·

## **COMMUNITY & EDUCATIONAL ENGAGEMENT**

### **Qur'anic & Foundational Literacy Instructor**

- Taught foundational literacy and Qur'anic recitation to learners with different levels of prior knowledge and learning pace.
- Adapted explanations and instructional approaches to individual learning needs.

**HerDream Foundation** | Community Outreach Volunteer | Bimbilla, Ghana | 2022

- Participated in community outreach and education activities involving young people and community members.

### **Reusable Sanitary Pad Training & Community Education**

- Contributed to community health-education training including reusable sanitary pads for female orphans and outreach initiatives.

**Member, KNUST Writers Association**

## **TECHNICAL & RESEARCH SKILLS**

**Programming & Web:** Python, SQL, JavaScript, HTML, CSS, Flask, FastAPI, Next.js

**AI & Data:** Machine Learning, Predictive Modeling, Random Forest, Data Preprocessing, Model Evaluation

**Cybersecurity:** Kali Linux, Nmap, Burp Suite, Network Reconnaissance, Web Security, SQL Injection Testing, CTF Methodology

**Databases:** MongoDB, PostgreSQL, SQLAlchemy, Structured Data Processing

**Communication:** Technical Writing, Research and Report Writing, Teaching, Mentoring, Public Speaking

## **PROFESSIONAL TRAINING**

Google Cybersecurity Professional Certificate | Google

Google AI Fundamentals | Google

IBM AI Fundamentals | IBM

Introduction to Cybersecurity | Cisco

Networking Basics | Cisco

Certified Phishing Prevention specialist | Hack and Fix

Certified Cybersecurity Education professional | Redteamleaders  
Leadership & Public Speaking | International Leadership Foundation

## **LANGUAGES**

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English | Arabic | Dagbani | Hausa